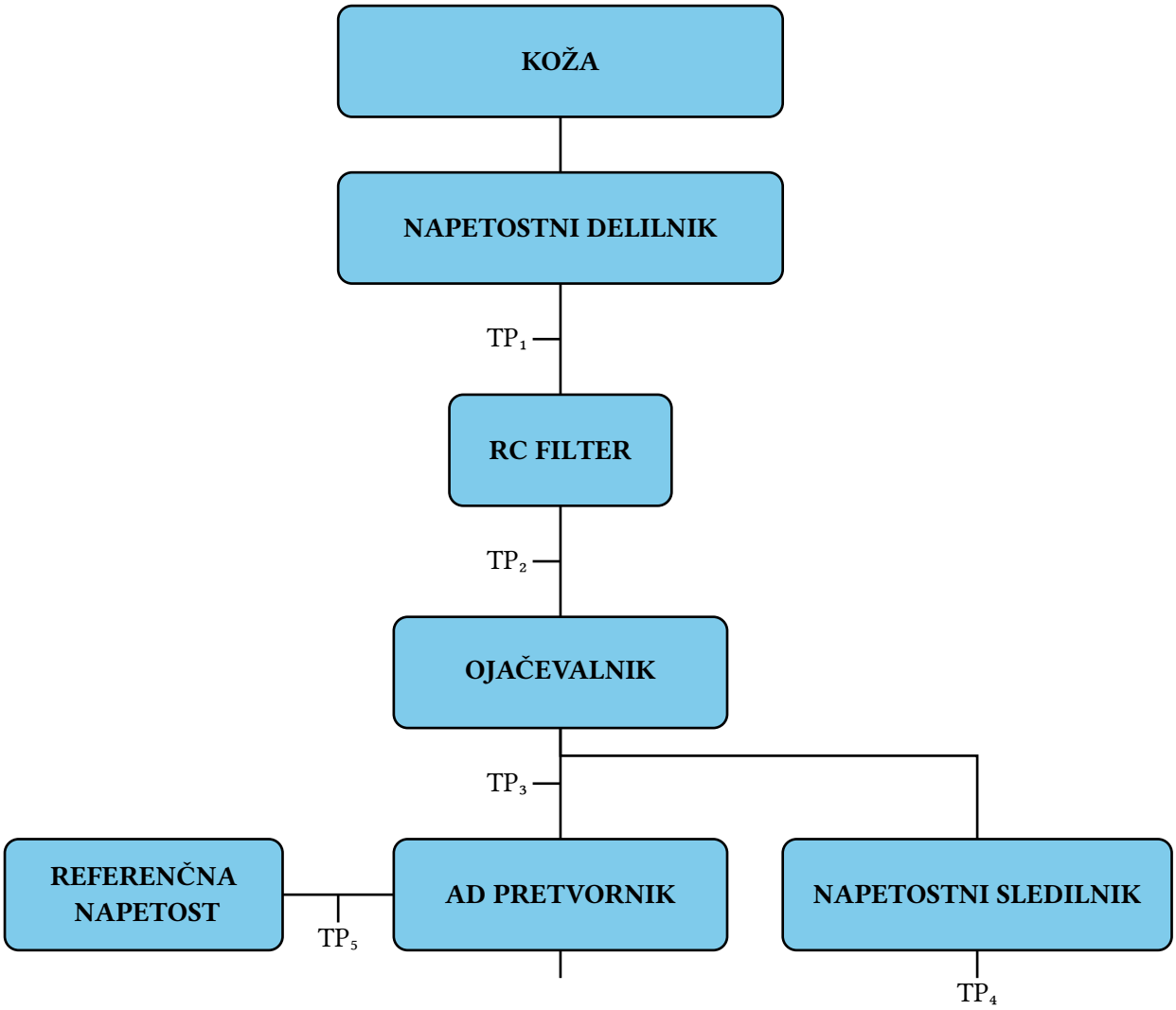
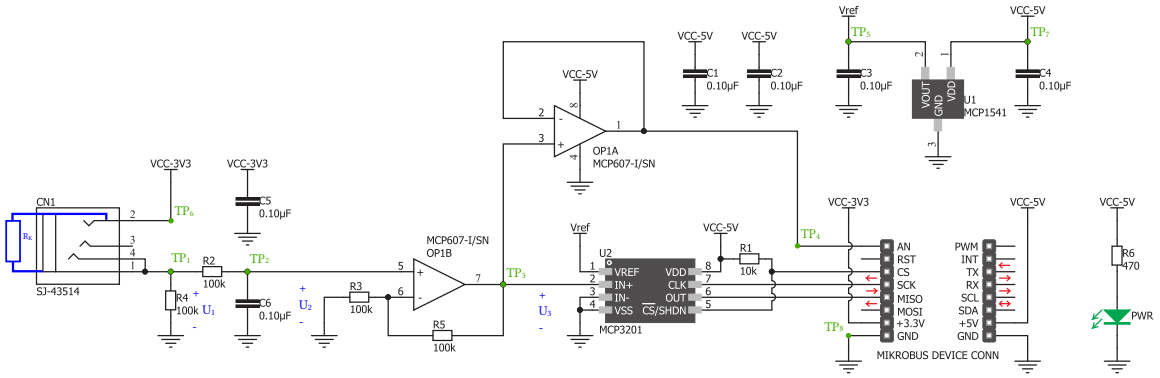
Projekt EDA (GSR Click) – Bločni diagram in tabela napovedi

Neža Povh, Vid Sojer

1 Bločni diagram



2 Shema z označenimi testnimi točkami

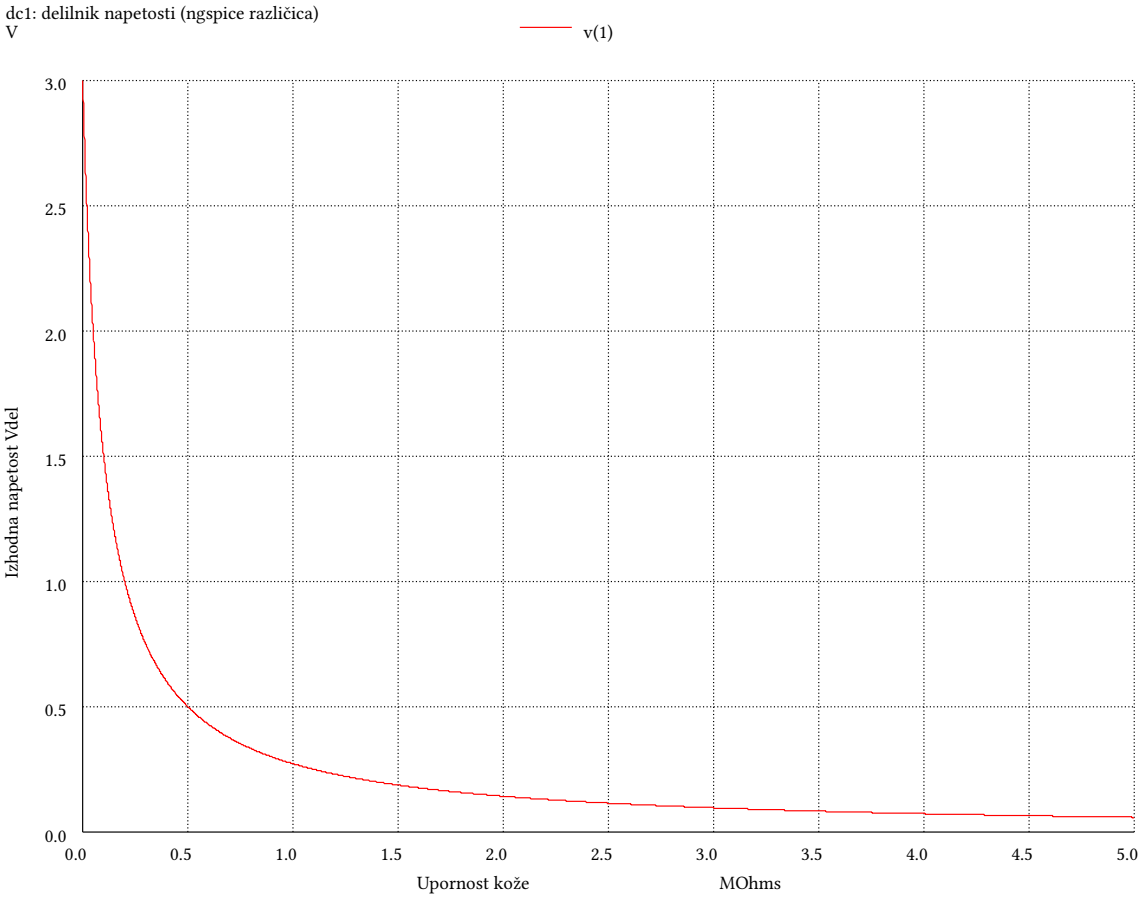


2.1 Napetostni delilnik

Enačba za izhodno napetost delilnika:

$$U_1 = U_{\text{ref del}} * \frac{R_4}{R_K + R_4}$$

Odvisnost U_1 od R_K:

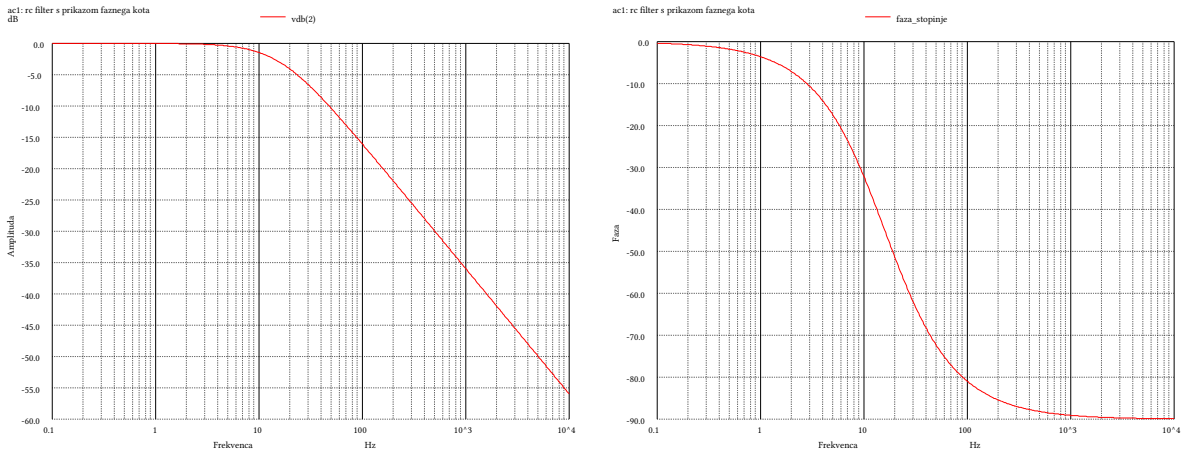


2.2 RC filter

Izračun mejne frekvence:

$$f_{\mathrm{m}} = \frac{1}{2\pi R_2 C_6} = \frac{1}{2\pi * 100\,\mathrm{k\Omega} * 0{,}1\,\mu\mathrm{F}} \approx 15{,}9\,\mathrm{Hz}$$

Frekvenčni odziv filtra pri frekvencah 0,1 Hz do 10 kHz:



2.3 Ojačevalnik

Ojačenje:

$$A_{\mathrm{U}} = 1 + \frac{R_5}{R_3}$$

$$R_5 = R_3 \Rightarrow A_{\mathrm{U}} = 1 + 1 = 2$$

Izhodna napetost:

$$U_3 = U_2 * A_{\mathrm{U}} = U_2 * 2$$

Nasičenje:

$$U_3 = U_{\mathrm{nap}} = 5\ \mathrm{V} \Rightarrow$$

$$U_{3\ \mathrm{nas}} = \frac{U_{\mathrm{nap}}}{A_{\mathrm{U}}} = \frac{5\,\mathrm{V}}{2} = 2{,}5\,\mathrm{V}$$

2.4 A/D pretvornik

N vrednost:

$$12\ \mathrm{bitni} \Rightarrow N_{\mathrm{MAX}} = 2^{12} - 1 = 4095$$

$$N = \min\left(\left\lfloor \frac{U_3}{U_{\mathrm{ref}}} * 2^{12} \right\rfloor, 2^{12} - 1\right)$$

Izračun vhodne napetosti:

$$U_3 = \frac{N}{2^{12}} * U_{\mathrm{ref}}$$

Nasičenje:

$$U_3 \geq U_{\mathrm{ref}} = 4{,}096\,\mathrm{V}$$

2.5 Napetostni sledilnik

$$A_{\mathrm{U}} = 1$$
$$U_{\mathrm{AN}} = U_3$$

2.6 Referenčna napetost

Referenčna napetost, definirana v podatkovnem listu:

$$U_{\mathrm{ref}} = 4{,}096\ \mathrm{V}$$

3 Meje nasičenja

- Zgornja meja A/D pretvornika:

$$4095 \quad (U_3 = 4,096\ \mathrm{V})$$

$$U_2 = \frac{U_3}{2} = 2,048\ \mathrm{V} = U_1$$

$$U_1 = U_{\mathrm{CC}} \frac{R_4}{R_{\mathrm{K}} + R_4}$$
$$U_1(R_{\mathrm{K}} + R_4) = U_{\mathrm{CC}} R_4$$
$$R_{\mathrm{K}} = \frac{U_{\mathrm{CC}} R_4}{U_1} - R_4 = R_4 \left(\frac{U_{\mathrm{CC}}}{U_1} - 1 \right)$$

$$G_S = \frac{1}{R_{\mathrm{K}}} = 1,4\ \mu\mathrm{S} \quad \rightarrow \quad N = 4095$$

- MCP607

$$U_{\mathrm{SS}} - 0,3\ \mathrm{V} \leq U_{\mathrm{IN}} \leq U_{\mathrm{DD}} - 1,1\ \mathrm{V}$$
$$-0,3\ \mathrm{V} \leq U_3 \leq 3,9\ \mathrm{V}$$
$$U_1 = U_2 = \frac{U_3}{2} \quad \rightarrow \quad U_1 \leq 1,95\ \mathrm{V}$$

$$R_S = 69,2\ \mathrm{k\Omega}$$
$$G = 14,4\ \mu\mathrm{S} \quad (\text{manj od A/D})$$

- Spodnja meja A/D pretvornika:

$$N = 1$$
$$N = 4096 \frac{u_3}{U_{\mathrm{ref}}} \quad \rightarrow \quad U_3 = \frac{N * U_{\mathrm{ref}}}{4096} = 1\ \mathrm{mV}$$
$$U_1 = 0,5\ \mathrm{mV}$$
$$R_S = R_4 \left(\frac{U_{\mathrm{CC}}}{U_1} - 1 \right)$$
$$G = 1,5\ \mathrm{nS}$$

4 Ločljivost

$$
\begin{aligned}
N &= x\frac{R_4 G}{1 + R_4 G} \\
N(1 + R_4 G) &= x R_4 G \\
N + N R_4 G &= x R_4 G \\
N R_4 G - x R_4 G &= -N \\
G(N R_4 - x R_4) &= -N \\
G &= \frac{N}{R_4 (x - N)}
\end{aligned}
$$

$$
\begin{aligned}
\frac{dG}{dN} &= \frac{R_4(x-N) - N(-R_4)}{\left(R_4(x-N)\right)^2} \\
&= \frac{R_4 x - N R_4 + N R_4}{R_4^2 (x-N)^2} \\
&= \frac{x}{R_4 (x-N)^2}
\end{aligned}
$$

$$
\begin{aligned}
\text{nizek G (N} \to 0): \quad \frac{dG}{dN} &= 1{,}2\ \text{nS} \\
\text{visok G (N} = 4095): \quad \frac{dG}{dN} &= 0{,}0105\ \text{nS}
\end{aligned}
$$

5 Največji tok skozi kožo

$$
I = \frac{U_{\mathrm{CC}}}{R_{\mathrm{K}} + R_4}
$$

$$
R_{\mathrm{K}} \to 0 \Rightarrow I < \frac{U_{\mathrm{CC}}}{R_4} = 33\,\mu\text{A}
$$

R_K	I
0 kΩ	33,0 µA
100 kΩ	16,5 µA
1000 kΩ	3,0 µA
∞	0

6 Tabela napovedanih vrednosti

G_K	R_K	TP_1	TP_2	TP_3	TP_4	TP_5	TP_6	TP_7	TP_8	N_{ADC}
0,00 µS	∞	0,000 V	0,000 V	0,000 V	0,000 V	4,096 V	3,3 V	5,0 V	0 V	0
21,27 µS	47 kΩ	2,245 V	2,245 V	4,490 V	> 3,9 V	4,096 V	3,3 V	5,0 V	0 V	4095
14,70 µS	68 kΩ	1,964 V	1,964 V	3,929 V	> 3,9 V	4,096 V	3,3 V	5,0 V	0 V	3928
10,00 µS	100 kΩ	1,650 V	1,650 V	3,300 V	3,300 V	4,096 V	3,3 V	5,0 V	0 V	3300
4,54 µS	220 kΩ	1,031 V	1,031 V	2,063 V	2,063 V	4,096 V	3,3 V	5,0 V	0 V	2062
2,12 µS	470 kΩ	0,579 V	0,579 V	1,158 V	1,158 V	4,096 V	3,3 V	5,0 V	0 V	1157
1,00 µS	1000 kΩ	0,300 V	0,300 V	0,600 V	0,600 V	4,096 V	3,3 V	5,0 V	0 V	600
0,45 µS	2200 kΩ	0,143 V	0,143 V	0,287 V	0,287 V	4,096 V	3,3 V	5,0 V	0 V	286
0,21 µS	4700 kΩ	0,069 V	0,069 V	0,137 V	0,137 V	4,096 V	3,3 V	5,0 V	0 V	137